

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	G. Jaswanth	Reg. No.:	192472235
Section / Batch:	2024	Date:	16/07/2026

Code of Conduct

I G. Jaswanth - 192472235 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Jaswanth

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

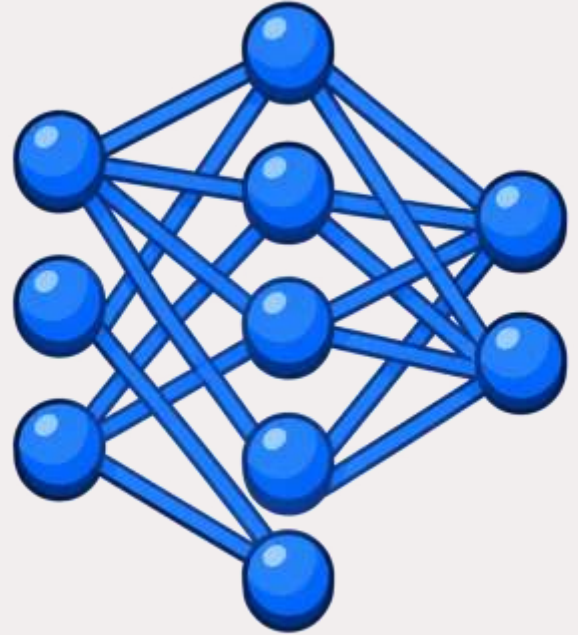
Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Language Models in Natural Language Processing (NLP)

Name : G. Jaswanth

RegNo : 192472235





What is a Language Model?



Predicting Word Sequences

A system that learns the **probability of word sequences** — predicting what comes next in a sentence.



Everyday Analogy

Your phone's keyboard suggests the next word as you type — that's a language model in action.



Trained on Massive Data

Captures grammar, facts, and context by learning from enormous collections of text.

Why Do We Need Language Models?

Ambiguous by Nature

Human language is **ambiguous, contextual, and constantly evolving** — impossible to hard-code.

Classic Example

"I saw the man with the telescope" — who has the telescope? Context decides.

Handling Scale

Language models help machines resolve ambiguity across **billions of daily interactions**: search, translation, voice assistants.



How Language Models Work



Each piece of text is broken into tokens, converted to numerical vectors, then fed through a model that predicts the next token — repeating millions of times during training to minimize errors.

Types of Language Models

Type	Key Idea	Example
N-gram	Count word co-occurrences	Spell checkers
RNN / LSTM	Sequential memory	Early chatbots
Transformer	Self-attention mechanism	GPT-4, BERT

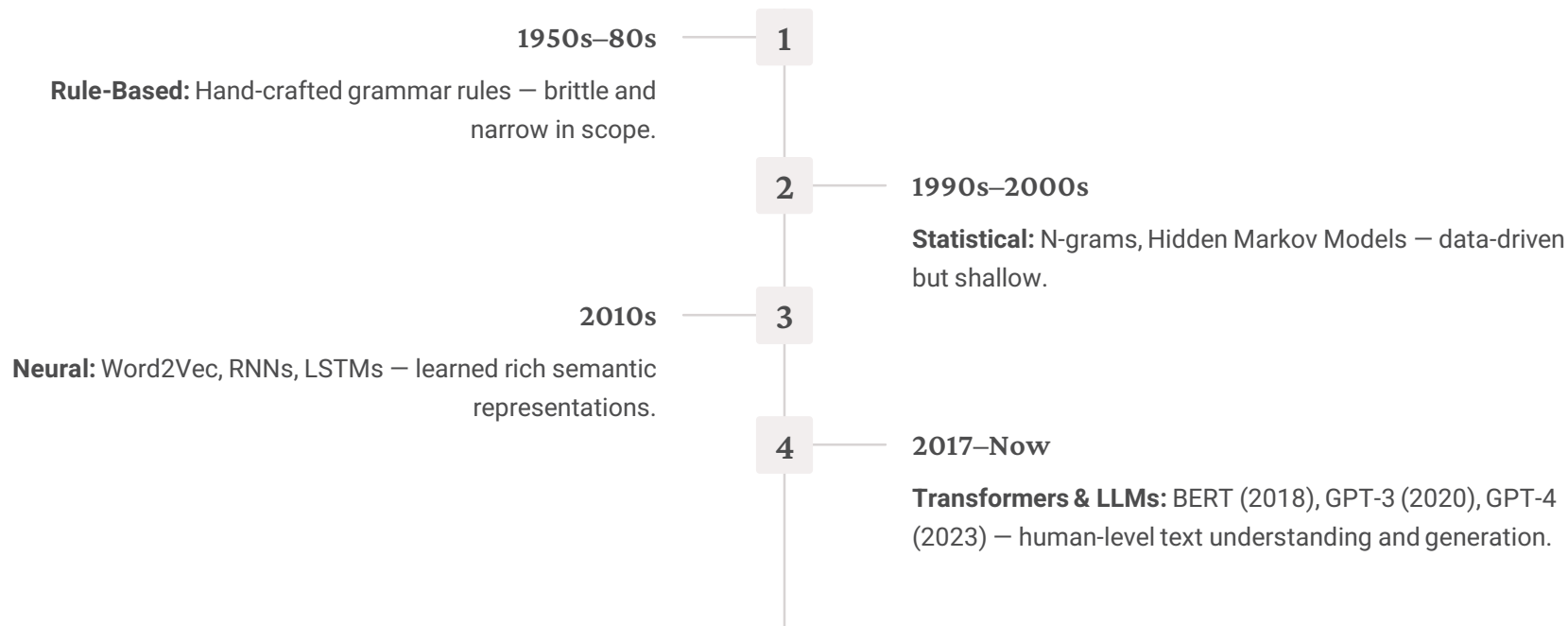
From Simple Counts to Deep Attention

N-gram models predict the next word from just the previous N words — fast but forgetful.

RNNs and LSTMs introduced sequential memory, allowing context to persist over longer text passages.

Transformers revolutionized the field by attending to **all words simultaneously** — today's gold standard powering GPT-4 and BERT.

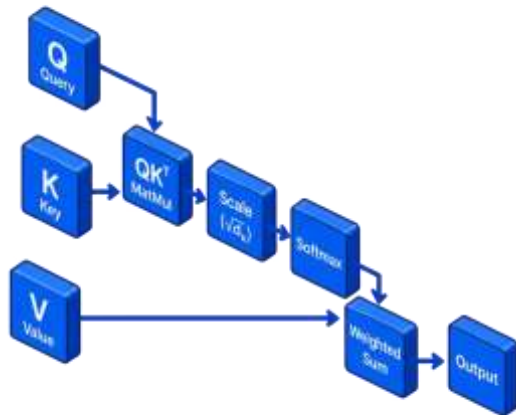
Evolution of Language Models



Language modeling has progressed from hand-crafted statistical models to large language models capable of processing, understanding, and generating human-level text at remarkable scale.

The Transformer: The Game-Changer

Introduced in "**Attention Is All You Need**" (Vaswani et al., 2017)



Self-Attention

Every word looks at every other word simultaneously to understand context.

Resolves Ambiguity

"Bank by the river"
vs. "Money in the bank" – attention knows the difference.

Parallel Training

Enables training on billions of parameters in parallel – the foundation of GPT & BERT.

Real-World Applications



Search Engines

Google uses BERT to understand the true intent behind your queries.



Machine Translation

Google Translate handles 100+ languages in real time for billions of users.



Chatbots & Assistants

ChatGPT, Siri, Alexa, and customer support bots all rely on language models.



Healthcare

Extracting insights from clinical notes and medical literature at scale.



Advantages & Challenges

✓ Strengths

- Fluent, coherent text generation
- Multilingual capabilities
- Few-shot learning across tasks
- Broad task coverage without retraining

⚠ Hallucinations

Models confidently generate false information – a critical reliability concern.

⚠ Bias

Training data reflects societal biases – outputs can be unfair or harmful.

⚠ Cost & Scale

GPT-3 training cost ~\$4.6M; acquiring general-purpose language ability requires training billions of parameters on massive text data, as predicted by scaling laws.

The Future of Language Models



Multimodal AI

Models that understand text, images, audio, and video together — GPT-4o leads the way.



Smaller, Efficient Models

Running powerful LLMs on phones and edge devices — bringing AI to everyone, everywhere.



Reasoning & Agents

LLMs that plan, use tools, and take autonomous actions — moving beyond text generation.



Responsible AI

Alignment, explainability, and reducing hallucinations are the top priorities for the field.

LLMs like ChatGPT have shown impressive capabilities in NLP tasks — and a systematic exploration of their full potential across the field is only just beginning.



Key Takeaways

01

Core Principle

Language models learn to **predict and generate text** from vast amounts of training data.

02

The Arc of Progress

Rule-Based → **Statistical** → **Neural** → **Transformers** → **LLMs** — each era unlocking new capabilities.

03

Already Around You

They power tools you use every day — search, translation, voice assistants, and more.

04

Exciting but Unfinished

An exciting future lies ahead — but challenges like **bias and hallucination** must be solved responsibly.

$$P(\text{she} | \text{The girl}) = \frac{0}{3} = 0$$

$$P(\text{boy} | \text{The girl}) = \frac{0}{3} = 0$$

$$P(\text{cat} | \text{The girl}) = \frac{0}{3} = 0$$

$$P(\text{boy} | \text{The girl}) = \frac{0}{2} = 0$$

$$P(\text{played} | \text{The girl}) = \frac{1}{2} = 0.5$$

$$P(\text{who} | \text{The girl}) = \frac{0}{2} = 0$$

Smoothing 0-infinity values
 Laplace / interpolation
 let $p = 10^{-6}$
 let $p = 10^{-6}$ again

stic Language Model

→ predict the next word

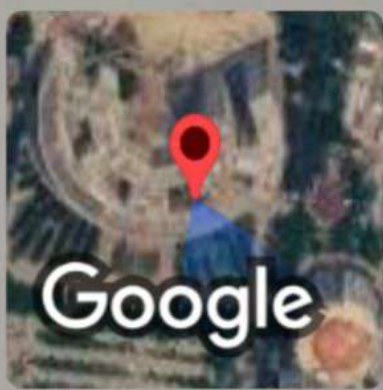
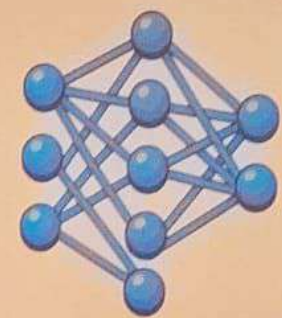
mas

Sequence of a number of words

am learning NLP

Language Models in Natural Language Processing (NLP)

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